

**GRAPH THEORY:
HOMEWORK 3
BOOK SECTION 1.3.**

ROB TOMPKINS

1. Determine the maximum number of edges in a bipartite subgraph of P_n , C_n , and K_n . (Hint: for K_n it may be helpful to review the exercise from §1.1 on counting edges of bipartite graphs.)

Solution. □

2. Prove that an even graph has no cut-edge

Proof. □

3. The problem involves the hypercube Q_k .

- (a) Use the recursive description of Q_k (Example 1.3.8) to prove that $|e(Q_k)| = k2^{k-1}$
- (b) Prove that $K_{2,3}$ is not contained in any hypercube Q_k . (Hint: prove that two distinct vertices in Q_k cannot have three common neighbors.)

Proof. □

4. Prove that every edge in the Petersen graph G belongs to exactly four 5-cycles, and use this to show that the Petersen graph has exactly twelve 5-cycles. (Hint: For the first part, extend the edge to a copy of P_4 , and apply Proposition 1.1.38.)

Proof. □

5. Use complete graphs and counting arguments (not algebra!) to prove that

- (a) $\binom{n}{2} = \binom{k}{2} + k(n-k) + \binom{n-k}{2}$ for all integers $0 \leq k \leq n$. (Hint: Count the edges of the complete graph K_n in two different ways.)
- (b) If $\sum n_i = n$ then $\sum \binom{n_i}{2} \leq \binom{n}{2}$

Proof. □